

Invariant measures for the stochastic heat equation on $\mathbb{R}^d$ without drift term

Le Chen
Auburn University
le.chen@auburn.edu

Nicholas Eisenberg
Auburn University
nze0019@auburn.edu

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Abstract

This paper deals with the long term behavior of the solution to the nonlinear stochastic heat equation $\partial u / \partial t - \frac{1}{2} \Delta u = b(u) \dot{W}$ for a Lipschitz continuous diffusion coefficient b which is not necessarily bounded. Using the theory of the stochastic integral laid out by John Walsh, we provide conditions which will guarantee the existence of an invariant measure for a broad range of initial conditions which includes L^2_ρ as well as the Dirac delta distribution δ_0 .

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1 Introduction

We consider the following *stochastic heat equation* (SHE):

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta u(t, x) = b(x, u(t, x)) \dot{W}(t, x) & x \in \mathbb{R}^d, t > 0, \\ u(0, \cdot) = \mu(\cdot), \end{cases} \quad (1.1) \quad \boxed{\text{E:SHE}}$$

The noise, $\dot{W}(t, s)$, is a centered Gaussian noise that is white in time and homogeneously colored in space defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with natural filtration $(\mathcal{F}_t)$:

$$\mathcal{F}_t = \sigma(W_s(A) : 0 \leq s \leq t, A \in \mathcal{B}_b(\mathbb{R}^d)) \vee \mathcal{N},$$

where $\mathcal{B}_b(\mathbb{R}^d)$ are the bounded Borel sets and $\mathcal{N}$ are the $\mathbb{P}$ -null sets of Ω . Its covariance structure, J , is defined on $C_c^\infty(\mathbb{R}^{d+1}) \times C_c^\infty(\mathbb{R}^{d+1})$ as follows:

$$J(\psi, \phi) := \mathbb{E} [\dot{W}(\psi) \dot{W}(\phi)] = \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx) (\psi(s, \cdot) * \tilde{\phi}(s, \cdot))(x), \quad \tilde{\phi}(x) := \phi(-x), \quad (1.2) \quad \boxed{\text{E:Cor}}$$

where Γ is a nonnegative and nonnegative definite tempered measure on $\mathbb{R}^d$ that is referred to as the *correlation measure*. The Fourier transform of Γ (in the generalized sense) is also a nonnegative and nonnegative definite tempered measure, which is called *spectral measure* and is denoted by $\hat{f}(d\xi)$. Moreover, in the case where Γ has density f , then $\hat{f}(d\xi) = \hat{f}(\xi) d\xi$.

Any jointly measurable $L^2(\Omega)$ -continuous processes, $u = (u(t, x) : t \geq 0, x \in \mathbb{R}^d)$, that is adapted to the natural filtration generated by W and solves almost surely the following stochastic integral equation is referred to as a *mild solution* to (1.1):

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x - y) \mu(dy) + \int_0^t \int_{\mathbb{R}^d} b(y, u(s, y)) G(t - s, x - y) W(ds, dy). \quad (1.3) \quad \boxed{\text{E:gmsol}}$$

In (1.3), $G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{|x|^2}{2t}\right)$ is the heat kernel and the stochastic integral is the *Walsh integral* (see [walsh:86:introduction; dalang.khoshnevisan.ea:09:minicourse]). For some constants $L_b > 0$ and $L_0 \geq 0$, the function $b(x, u)$ satisfies the following Lipschitz and boundedness conditions:

$$|b(x, u) - b(x, v)| < L_b |u - v| \quad \text{and} \quad |b(x, 0)| \leq L_0 \quad \text{for all } u, v \in \mathbb{R}, x \in \mathbb{R}^d. \quad (1.4) \quad \boxed{\text{E:lipcon}}$$

In particular, our assumption allows the linear case $b(x, u) = \lambda u$, which is usually referred to the *parabolic Anderson model* [carmona.molchanov:94:parabolic].

The aim of this paper is to investigate the conditions required to guarantee the existence of invariant measures for the solution to (1.1), which is a crucial step towards the study of the ergodicity of the system. We will see below in Theorem 1.3 that in our setting, the solution to (1.1) almost surely belongs to a weighted space of square integrable function, $L^2_\rho(\mathbb{R}^d)$. By an invariant measure for (1.1), we mean a measure, η , on the space $(E, \mathcal{B}(E))$ such that

$$\eta(A) = \int_E \mathcal{L}(u(t, \cdot, \zeta))(A) \eta(d\zeta), \quad \forall t \geq 0,$$

where $A \in \mathcal{B}(E)$ and $\mathcal{L}(u(t, \cdot, \zeta))(A)$ is the law of (1.1) starting from ζ . This is the only part left of the paper that I want to confirm. I think E needs to be the bounded space of continuous functions. This is how it is defined in [da-prato.g-atarek.ea:92:invariant] and [assing.manthey:03:invariant]. How do we define the laws for random initial conditions, φ , that are almost surely in $L^2_\rho(\mathbb{R}^d)$? See (1.10). I believe there may not be an issue because of the $\varphi \in \mathcal{F}_0$ assumption but I want to make sure. It seems that other authors ignore this issue.

After reading [assing.manthey:03:invariant] a little more carefully, I think what we need as our definition is

$$\eta(A) = \int_E \mathcal{L}(u(t, \cdot, \zeta))(A) \eta(d\zeta), \quad \forall t \geq t_0,$$

where ζ is any measure satisfying (1.9). I believe that η would still be a measure on the space $L^2_\rho(\mathbb{R}^d)$. Thus we have expanded the domain of

$$\zeta \mapsto \mathcal{L}(u(t, \cdot, \zeta))(A), \quad t \geq t_0,$$

from $\zeta \in C_b(\mathbb{R}^d)$ to $\zeta \in M$ where M is the space of deterministic measures as in (1.9). Still unclear on all of this.

Existence in unbounded domains has been studied in few papers such as [assing.manthey:03:invariant; misiats.stanzhytskyi.ea:20:invariant; misiats.stanzhytskyi.ea:16:existence; tessitore.zabczyk:98:inv]. Since the spatial domain is unbounded, one needs to introduce a weight function ρ and the corresponding weighted space $L^2_\rho(\mathbb{R}^d)$ as usual; see Section 2 below for more details. Following the standard procedure as laid out by Da Prato and Zabczyk [da-prato.g-atarek.ea:92:invariant], the main effort is to show that:

$$\sup_{t>0} \mathbb{E} \left(\|u(t, \cdot)\|_{L^2_\rho(\mathbb{R}^d)}^2 \right) < \infty. \quad (1.5) \quad \boxed{\text{E:BddMnt}}$$

However, this solution is usually *intermittent*, namely, the probability moments to (1.1) have a certain exponential growth in t . In [assing.manthey:03:invariant; misiats.stanzhytskyi.ea:20:invariant; misiats.stanzhytskyi.ea:16:existence], a drift term $f(x, u)$ is included in (1.1) in a crucial way to help to cancel the otherwise exponential growing moments. The absence of such a drift term in this paper makes the problem more challenging. In order to have moments bounded in time, as required in (1.5), one has to first identify the exact conditions, under which the moments to (1.1) do not possess exponential growth. This question has been answered in [chen.kim:19:nonlinear], where necessary and sufficient conditions are given. More precisely, one has to have L_b (the Lyapunov exponent of b) in (1.4) small enough, the spatial dimension $d \geq 3$, and we must strengthen (2.6) to the following:

$$\Upsilon(0) < \infty. \quad (1.6) \quad \boxed{\text{E:U0finite}}$$

In the setting of equation (1.1), Tessitore and Zabczyk [tessitore.zabczyk:98:invariant] prove existence of an Invariant measure when starting from a deterministic $L^2_\rho(\mathbb{R}^d)$ function under the assumption the solution satisfies a boundedness in probability condition and also that the spectral density $\hat{f}$ satisfies

$$\hat{f} \in L^p(\mathbb{R}^d) \quad \text{where} \quad \frac{d-2}{d} < \frac{1}{p}; \quad (1.7) \quad \boxed{\text{E:CorTZ}}$$

see Hypothesis 2.1 (*ibid.*). However as was illustrated in [tessitore.zabczyk:98:invariant], the only concrete example given was for (1.1) starting from the constant 1 and the following additional assumptions were needed: $d \geq 3$ and the Lipschitz constant of b must satisfy $L(b)^{-2} > 1$, where

$$\tilde{\Gamma} := \left| \sqrt{\mathcal{F}(f)} \right| * \left| \sqrt{\mathcal{F}(f)} \right| \quad \text{and} \quad 1 := \frac{\Gamma(d/2 - 1)}{4\pi^{d/2}} \int_{\mathbb{R}^d} \tilde{\Gamma}(d\zeta) |\zeta|^{2-d}, \quad (1.8) \quad \boxed{\text{E:TZspec}}$$

where Γ is the Gamma function.

These additional assumptions given by Tessitore and Zabczyk parallel our assumptions (1.6) and (1.16), however one can see that both (1.6) and (1.16) are more compact and easier to check than (1.8). Moreover, our method allows us expand our choice of initial condition $\mu(\cdot)$. Some applicable deterministic initial measures include (see Example 1.5 below):

Dirac delta measure $\delta_0(dx)$, Lebesgue measure dx , $\varphi(x)dx$ with $\varphi \in C^\infty(\mathbb{R}^d)$,

but more generally, any initial data μ that satisfy (1.19) and (1.21) below, as well as one of the following:

$$(1) \mu(dx) \text{ is a deterministic, locally finite, regular, signed Borel measure with the corresponding Hahn decomposition } \mu = \mu_+ - \mu_- \quad (1.9) \quad \boxed{\text{E:DetInit}}$$

$$(2) \mu(dx) = \varphi(x)dx \text{ where } \varphi \text{ is an } \mathcal{F}_0 \text{ measurable process in } L^2(\Omega, L^2_\rho(\mathbb{R}^d)), \text{ namely } \mathbb{E} \left(\|\varphi(\cdot)\|_{L^2_\rho(\mathbb{R}^d)} \right) < \infty. \quad (1.10) \quad \boxed{\text{E:RandInit}}$$

Remark 1.1. We point out that (1.1) starting from any deterministic measure, μ , as in (1.9) can be viewed as a solution of (1.1) starting from a process $\varphi \in L^2(\Omega, L^2_\rho(\mathbb{R}^d))$. Indeed, let u denote the mild solution to (1.1) starting from μ which satisfies the conditions of Theorem 1.4. For any $t_0 > 0$, consider the following restarted equation:

$$\begin{cases} \frac{\partial v}{\partial t}(t, x) - \frac{1}{2} \Delta v(t, x) = b(x, v(t, x)) \dot{W}_{t_0}(t, x) & x \in \mathbb{R}^d, t > 0, \\ v(0, \cdot) = u(t_0, \cdot), \end{cases} \quad (1.11) \quad \boxed{\text{E:restart}}$$

where $\dot{W}_{t_0}(t, x) := \dot{W}(t + t_0, x)$ denotes the time shifted noise. Ie,

$$\int_0^t \int_{\mathbb{R}^d} W_{t_0}(ds, dy) = \int_{t_0}^{t+t_0} \int_{\mathbb{R}^d} W(ds, dy).$$

By Theorem 2.5, $u(t_0, \cdot)$ is $\mathcal{F}_{t_0}$ measurable and by (1.17), we see that $u(t_0, \cdot) \in L^2(\Omega, L^2_\rho(\mathbb{R}^d))$. Moreover, the semi-group property of the heat kernel implies that for any $t \geq 0$, we have $u(t + t_0, x) \stackrel{\mathcal{L}}{=} v(t, x)$ where $\mathcal{L}$ denotes equality in law. Since we are interested in studying the long time behavior of (1.1), we obtain the same information by choosing to study the restarted equation. For this reason unless we state otherwise, we will focus our attention completely on (1.1) starting from a $\mathcal{F}_0$ measurable processes $\varphi \in L^2(\Omega, L^2_\rho(\mathbb{R}^d))$, in others words (1.1) starting from data which satisfy (1.10). This simplification of (1.1) to the restarted equation is key in order to argue the existence of an invariant measure. See Example 1.5 below for a continue of this discussion.

We now give some notation.

Notation 1.2. We will use $\|\cdot\|_\rho$ instead of $\|\cdot\|_{L^2_\rho(\mathbb{R}^d)}$. Now let $p \geq 1$. By defining $\|X\|_p := (\mathbb{E}(|X|^p))^{1/p}$,

$$J_0(t, x) := (G(t, \cdot) * \mu)(x) = \int_{\mathbb{R}^d} G(t, x - y) \mu(dy) \quad \text{and} \quad J_p(t, x) := (G(t, \cdot) * \|\mu\|_p)(x), \quad (1.12) \quad \boxed{\text{E:J}}$$

where we use the following convention:

$$\mu(dx) = \begin{cases} \mu(dx) & \text{when } \mu \text{ is deterministic} \\ \varphi(x)dx & \text{when } \mu \text{ is random,} \end{cases} \quad \|\mu(dx)\|_p = \begin{cases} |\mu|(dx) & \text{when } \mu \text{ is deterministic} \\ \|\varphi(x)\|_p dx & \text{when } \mu \text{ is random.} \end{cases}$$

Lastly, we denote

$$\mu_p^* := 1 + \|\mu\|_p \quad \text{and} \quad J_p^*(r, z) = (G(r, \cdot) * \mu_p^*)(x). \quad (1.13) \quad \boxed{\text{E:qJ*}}$$

We remind the reader that for deterministic initial measures

$$\mu_p^* = \mu^* = 1 + |\mu| \quad \text{and} \quad J_p^*(r, z) = J^*(r, z) = (G(r, \cdot) * \mu^*)(x). \quad (1.14) \quad \boxed{\text{E:J*}}$$

Again, whether μ is random or deterministic, we will always use the notation given in (1.13).

We may now a preliminary result used to prove Theorem (1.4) below.

T:Lrho

Theorem 1.3. Consider (1.1) starting from μ which satisfies (1.10). Assume that

(i) $\rho : \mathbb{R}^d \rightarrow \mathbb{R}_+$ is an nonnegative $L^1(\mathbb{R}^d)$ function;

(ii) the initial measure μ , either random or deterministic, satisfies that

$$\mathcal{G}_{\rho,2}(t) = \mathcal{G}_{\rho,\mu_2^*}(t) := \int_{\mathbb{R}^d} (J_2^*(t, x))^2 \rho(x) dx < \infty \quad \text{for all } t > 0, \quad (1.15) \quad \boxed{\text{E:G}}$$

(iii) the correlation measure Γ satisfies (1.6),

(iv) the Lipschitz constant L_b of b (see Theorem 2.6 below) satisfies

$$2L_b^2 < \frac{1}{128\Upsilon(0)}. \quad (1.16) \quad \boxed{\text{E:LipLb}}$$

Then for all $t > 0$, $u(t, \cdot) \in L^2_\rho(\mathbb{R}^d)$ a.s. Moreover, for some constant $C > 0$ which does not depend on t , it holds that

$$\mathbb{E} \left(\|u(t, \cdot)\|_\rho^2 \right) \leq C \mathcal{G}_{\rho,2}(t) < \infty. \quad (1.17) \quad \text{E:uInLrho}$$

Proof. The constant C may change from line to line. Under conditions (iii) and (iv), we can apply Fubini's Theorem and the moment bound (2.9) to see that there exists a constant C independent of t such that,

$$\begin{aligned} \mathbb{E} \left(\|u(t, \cdot)\|_\rho^2 \right) &= \mathbb{E} \left(\int_{\mathbb{R}^d} |u(t, x)|^2 \rho(x) dx \right) \\ &\leq C \int_{\mathbb{R}^d} \left(1 + (G(t, \cdot) * \|\mu\|_2)(x) \right)^2 \rho(x) dx \\ &= C \int_{\mathbb{R}^d} \left((G(t, \cdot) * (1 + \|\mu\|_2))(x) \right)^2 \rho(x) dx \\ &= C \mathcal{G}_{\rho,2}(t) < \infty \end{aligned}$$

This proves Theorem 1.3. □

Here is the main theorem of the paper:

T:Main

Theorem 1.4. Consider (1.1) starting from μ which satisfies (1.10). Assume that all assumptions of Theorem 1.3 hold. Let $q > 2$ be arbitrary but fixed. Moreover, suppose that

(i) if ρ is an admissible weight and there exists another admissible weight $\hat{\rho}$ such that

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\hat{\rho}(x)} dx < \infty, \quad (1.18) \quad \text{E:rhorho}$$

(ii) the weight function $\hat{\rho}$ and the initial condition satisfy the following condition:

$$\limsup_{t>0} \mathcal{G}_{\hat{\rho},q}(t) < \infty, \quad (1.19) \quad \text{E:InitData}$$

(iii) for some $\alpha \in (0, 1/2)$, the spectral measure $\hat{f}$ satisfies that

$$\mathcal{H}_\alpha(t) := \int_0^t dr r^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-r|\xi|^2) < \infty \quad \text{for all } t > 0, \quad (1.20) \quad \text{E:H}$$

(iv) for $\alpha \in (q^{-1}, 2^{-1})$,

$$\int_0^t [\mathcal{G}_{\hat{\rho},q}(s) \mathcal{H}_\alpha(s)]^{q/2} ds < \infty \quad \text{for all } t > 0, \quad (1.21) \quad \text{E:GH-int}$$

(v) the condition (1.16) is strengthened to

$$qL_b^2 < \frac{1}{128\Upsilon(0)}.$$

Then we have that

(a) for any $t_0 > 0$, the sequence of laws of $\{\mathcal{L}u(t, \cdot)\}_{t \geq t_0}$ is tight, i.e., for any $\epsilon \in (0, 1)$, there exists a compact set $\mathcal{K} \subset L^2_\rho(\mathbb{R}^d)$ such that

$$\mathcal{L}u(t, \cdot)(\mathcal{K}) := \mathbb{P}(u(t, \cdot) \in \mathcal{K}) \geq 1 - \epsilon, \quad \text{for all } t \geq t_0; \quad (1.22) \quad \text{E:cpt}$$

(b) there exists an invariant measure for the laws $\{\mathcal{L}u(t, \cdot)\}_{t>0}$.

Proof of Theorem 1.4 (b). Fix an arbitrary $T_0 > 0$ and denote

$$U(T) := \frac{1}{T} \int_{T_0}^{T+T_0} \mathcal{L}(u(t, \cdot)) \, dt, \quad T > 0.$$

We claim that the family of laws $U(T, \cdot)$ for $T > 0$ is tight in $L^2_\rho(\mathbb{R}^d)$. Indeed, for any $\epsilon \in (0, 1)$, by part (a), there exists a compact set $\mathcal{K} \in L^2_\rho(\mathbb{R}^d)$ such that (1.22) holds. This implies that

$$U(T)(\mathcal{K}^c) = \frac{1}{T} \int_{T_0}^{T+T_0} \mathcal{L}(u(t, \cdot))(\mathcal{K}^c) \, dt \leq \frac{1}{T} \int_{T_0}^{T+T_0} \epsilon \, dt = \epsilon, \quad \text{for all } T > 0.$$

Let $\{T_n\}_{n \in \mathbb{N}}$ be any deterministic sequence such that $T_n \uparrow \infty$. Since $\{U(T_n)\}_{n \geq 1}$ is a tight sequence of measures, there exists a subsequence $\{U(T_{n_m})\}_{m \geq 1}$ that converges weakly to a measure, say, η , on $L^2_\rho(\mathbb{R}^d)$. Then one can apply for the *Krylov-Bogoliubov theorem* (see, e.g., [da-prato.zabczyk:96:ergodicity]) to conclude that the measure η is the invariant measure for $\mathcal{L}(u(t, \cdot))$, $t \geq T_0$. Finally, we can choose T_0 to be arbitrarily close to zero. This completes the proof of part (c) of Theorem 1.4. $\square$

E:Gamma+f

Example 1.5 (Bessel kernel and δ_0 initial data). We end this section with an example of an applicable initial data and spectral density. Note that because of (1.21), the initial data and spectral density must exist in tandem. Let's consider the initial data $\mu(\cdot) = \delta_0(\cdot)$ and let the correlation function of W be $f(x) = B_s(x)$, where B_s denotes the Bessel kernel with parameter s . It is shown in [grafakos:14:modern] that there exists $c(s, d)$ and $C(s, d)$ such that for $0 < s < d$

$$B_s(x) \leq C(s, d) \exp(-|x|/2) \quad \text{for } |x| \geq 2,$$

and for $H(x) = |x|^{s-d} + 1 + O(|x|^{s-d+2})$,

$$\frac{1}{c(s, d)} \leq \frac{B_s(x)}{H_s(x)} \leq c(s, d) \quad \text{for } |x| \leq 2.$$

Hence $B_s(x)$ has blow up at the origin and exponential decay at infinity. Moreover,

$$\mathcal{F}G_s(\xi) = \frac{1}{(1 + |\xi|^2)^{s/2}}. \quad (1.23)$$

E:FB

It is easy to see that δ_0 satisfies (2.5). Moreover, it is easy to see that if $d - 2 < s < d$ then $\Upsilon(0) < \infty$ and so the conditions of Theorem (2.5) and Theorem 2.6 are satisfied and a unique mild solution, u , exists that starts from δ_0 and is driven by a Gaussian noise with correlation function $B_s(x)$. We denote $v(t, x)$ to be the restarted solution as in (1.11). We now show that v satisfies the conditions in Theorem 1.4. In other words, it remains to prove that $v(0, \cdot) = u(t_0, \cdot)$ and $B_s(x)$ satisfy the conditions (1.19), (1.20) and (1.21).

Let $q > 2$ be arbitrary. Starting with (1.19),

$$\begin{aligned} \mathcal{G}_{\hat{\rho}, q}(s) &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(s, x - y) \mu_q^*(dy) \right)^2 \rho(x) dx \\ &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(s, x - y) (1 + \|v(0, y)\|_q)(dy) \right)^2 \hat{\rho}(x) dx \\ &= \int_{\mathbb{R}^d} \left(1 + \int_{\mathbb{R}^d} G(s, x - y) \|v(0, y)\|_q dy \right)^2 \hat{\rho}(x) dx \end{aligned}$$

$$\leq 2 \|\widehat{\rho}\|_{L^1(\mathbb{R}^d)} + 2 \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(s, x-y) \|u(t_0, y)\|_q dy \right)^2 \widehat{\rho}(x) dx,$$

where in the last line we used Recall that $u(t_0, x) = v(0, x)$. In the following, C is a generic constant independent of s that may change from line to line. By applying the moment bound (2.9) we see that

$$\begin{aligned} \mathcal{G}_{\widehat{\rho}, q}(s) &\leq 2 \|\widehat{\rho}\|_{L^1(\mathbb{R}^d)} + 2C \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(s, x-y) [1 + (G(t_0, \cdot) * \delta_0)(y)] dy \right)^2 \widehat{\rho}(x) dx \\ &= 2 \|\widehat{\rho}\|_{L^1(\mathbb{R}^d)} + C \int_{\mathbb{R}^d} \left(1 + \int_{\mathbb{R}^d} G(s, x-y) (G(t_0, \cdot) * \delta_0)(y) dy \right)^2 \widehat{\rho}(x) dx \\ &= 2 \|\widehat{\rho}\|_{L^1(\mathbb{R}^d)} + C \int_{\mathbb{R}^d} (1 + (G(t_0 + s, \cdot) * \delta_0)(x))^2 \widehat{\rho}(x) dx \\ &\leq 2 \|\widehat{\rho}\|_{L^1(\mathbb{R}^d)} + 2C \|\widehat{\rho}\|_{L^1(\mathbb{R}^d)} + 2C \int_{\mathbb{R}^d} (G(t_0 + s, \cdot) * \delta_0)(x)^2 \widehat{\rho}(x) dx \\ &\leq C \|\widehat{\rho}\|_{L^1(\mathbb{R}^d)} + C \int_{\mathbb{R}^d} (G(t_0 + s, x))^2 \widehat{\rho}(x) dx \\ &= C \|\widehat{\rho}\|_{L^1(\mathbb{R}^d)} + C \frac{(2\sqrt{\pi})^{-d}}{(t_0 + s)^{d/2}} \int_{\mathbb{R}^d} G\left(\frac{t_0 + s}{2}, x\right) \widehat{\rho}(x) dx \\ &=: \mathbf{G}(s), \end{aligned}$$

and this verifies (1.19) because $t_0 > 0$ is fixed. Moreover, this implies that for any $T > 0$, $\mathcal{G}_{\widehat{\rho}, q}(s)$ is bounded on $s \in [0, T]$. This will be needed below.

Now let's verify (1.20) for the Bessel kernel, B_s . By recalling (1.23), we have

$$\begin{aligned} \mathcal{H}_\alpha(s) &= \int_0^s dr r^{-2\alpha} \int_{\mathbb{R}^d} d\xi \frac{\exp(-r|\xi|^2)}{(1 + |\xi|^2)^{s/2}} \\ &= \sigma(\mathbb{S}^{d-1}) \int_0^t dr r^{-2\alpha} \int_0^\infty dz \frac{\exp(-rz^2)}{(1 + z^2)^{s/2}} z^{d-1}. \end{aligned}$$

Let's examine the $\int_0^\infty dz$ integral and apply the change of variables $y = z^2$. This yields,

$$\int_0^\infty dz \frac{\exp(-rz^2)}{(1 + z^2)^{s/2}} z^{d-1} = \frac{1}{2} \int_0^\infty dy e^{-ry} (1 + y)^{s/2} y^{\frac{d}{2}-1} dy = \frac{\Gamma(d/2)}{2} U\left(\frac{d}{2}, \frac{d+2-s}{2}, r\right),$$

where $U(\alpha, \beta, t)$ is the confluent hypergeometric U function (see [oliver.lozier.boisvert.clark:2010:nist]) and the last equality can be found in [oliver.lozier.boisvert.clark:2010:nist]. Thus, to show the finiteness of $\mathcal{H}_\alpha(s)$ for each $s > 0$, it suffices to show that the following integral is finite:

$$\int_0^s r^{-2\alpha} U\left(\frac{d}{2}, \frac{d+2-s}{2}, r\right) dr. \quad (1.24) \quad \boxed{\text{E:BesselHG}}$$

The behavior of $U(\alpha, \beta, t)$ around the origin is given in [oliver.lozier.boisvert.clark:2010:nist] and is given in cases on β , which in our case is $(d+2-s)/2$. Recall above that we are taking $d-2 < s < d$. For this reason, the only case in [oliver.lozier.boisvert.clark:2010:nist] that is applicable is $1 < (d+2-s)/2 < 2$ and in this case

$$r^{-2\alpha} U\left(\frac{d}{2}, \frac{d+2-s}{2}, r\right) = O\left(r^{1-2\alpha-(d+2-s)/2} + r^{-2\alpha} + r^{2-2\alpha-(d+2-s)/2}\right). \quad (1.25)$$

So in order (1.24) to be finite we need

$$\min\{1 - 2\alpha - (d + 2 - s)/2, -2\alpha, 2 - 2\alpha - (d + 2 - s)/2\} \geq -1.$$

Recall that we are assuming $\alpha \in (q^{-1}, 2^{-1})$ so $-2\alpha > -1$. Thus it suffices to set

$$1 - 2\alpha - (d + 2 - s)/2, -2\alpha \geq -1 \iff 2 - 4\alpha - d + s > 0.$$

This is indeed possible as $2 - 4\alpha > 0$ and again because $d - 2 < s < d$, one can take s sufficiently close to d so that $(2 - 4\alpha) - (d - s) > 0$. This completes the verification for (1.20).

Lastly we need to verify (1.21). But essentially all the work for this has been completed. The verification of (1.20) implies that $\mathcal{H}_\alpha(s)$ is continuous on $s \in [0, T]$ for any $T > 0$ and we already saw above that $\mathcal{G}_{\hat{\rho}, q}(s)$ was bounded on $[0, T]$. These two facts imply (1.21), thus allowing us to apply Theorem (1.4) and verify the existence of an invariant measure to (1.1) with initial data, $\mu(\cdot) = u(t_0, \cdot)$ and noise W with correlation function given to be the Bessel kernel B_s . Since we are only concerned with the long term behavior of the solution and because for all $t > 0$ we have $u(t + t_0, x) \stackrel{\mathcal{L}}{=} v(t, x)$, what we have actually done was verify the existence of an invariant measure for (1.1) starting from δ_0 with correlation function $B_s(x)$.

Example 1.6 (More examples). We would like to point out that the following initial data are also applicable when considering the correlation function to be the Bessel kernel $B_s(x)$ with $s < d$:

$$\text{Lebesgue measure } dx \quad \text{and} \quad \varphi(x)dx \quad \text{with } \varphi \in C^\infty(\mathbb{R}^d).$$

The interested reader may go through the same steps above to see that (1.19), (1.20) and (1.21) are all finite.

2 Preliminaries

S:Prelim

Let ρ be some nonnegative function on $\mathbb{R}^d$. We denote by $L_\rho^2(\mathbb{R}^d)$ the weighted Hilbert space with the inner product $\langle f, g \rangle_\rho := \int_{\mathbb{R}^d} f(x)g(x)\rho(x) dx$ and set $\|f\|_\rho^2 := \int_{\mathbb{R}^d} f(x)^2\rho(x) dx$. Following Tessitore and Zabczyk [tessitore.zabczyk:98:invariant], we make the following definition:

D:Rho

Definition 2.1. $\rho : \mathbb{R}^d \mapsto \mathbb{R}$ is called an *admissible weight* if ρ is a strictly positive, bounded and continuous function in $L^1(\mathbb{R}^d)$ such that for $T > 0$ there exists a constant $C_\rho(T)$ such that

$$(G(t, \cdot) * \rho(\cdot))(x) \leq C_\rho(T)\rho(x) \quad \text{for all } t \in [0, T] \text{ and } x \in \mathbb{R}^d, \quad (2.1) \quad \text{E:aw}$$

or equivalently,

$$\sup_{x \in \mathbb{R}^d, t \in [0, T]} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)} \leq C_\rho(T). \quad (2.2) \quad \text{E:awbd1}$$

Several examples of admissible weights are (see Example in Section 2 of [tessitore.zabczyk:98:invariant])

$$\begin{aligned} \rho(x) &= \exp(-a|x|), & a > 0 \quad \text{and} \\ \rho(x) &= (1 + |x|^a)^{-1}, & a > d. \end{aligned} \quad (2.3) \quad \text{E:admRho}$$

R:admRho

Remark 2.2. The smaller the weight function $\rho(\cdot)$ (not necessarily to be admissible here) is, the larger the space $L_\rho^2(\mathbb{R}^d)$ is. For example, one may choose ρ to be either a nonnegative function with compact support or the heat kernel itself $G(1, \cdot)$. In both cases, ρ is smaller than those in (2.3) (up to a constant). However, one can easily check that the admissible condition (2.1) excludes these two cases. It is much less obvious that the following weight function is not admissible:

$$\rho(x) = \exp(-a|x|^b), \quad x \in \mathbb{R}^d, \quad \text{with } a > 0 \text{ and } b \in (1, 2) \text{ fixed.}$$

P:weight

Proposition 2.3 (Proposition 2.1 of [tessitore.zabczyk:98:invariant]). *For any admissible weight ρ , the operators on $L_\rho^2(\mathbb{R}^d)$ defined by $\varphi \mapsto (G(t, \cdot) * \varphi(\cdot))(x)$ can be extended to a C_0 -semigroup on L_ρ^2 . Moreover, if $\widehat{\rho}$ is another admissible weight such that*

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\widehat{\rho}(x)} dx < \infty,$$

then for any $t < 0$, the operators defined above are compact from $L_\rho^2(\mathbb{R}^d)$ to $L_{\widehat{\rho}}^2(\mathbb{R}^d)$.

We need Lemma 2.4 to show that $\varphi \in L_\rho^2$ will satisfy (2.5).

L:RhoDom

Lemma 2.4. *For any admissible weight function ρ and $a > 0$, there exists a constant $K_{\rho,a} > 0$ such that*

$$\exp(-a|x|^2) \leq K_{\rho,a} \rho(x), \quad \text{for all } x \in \mathbb{R}^d. \quad (2.4) \quad \text{E:RhoDom}$$

Proof. Let ρ be an arbitrary admissible weight function and $a > 0$. By (2.1),

$$\begin{aligned} \frac{\exp(-a|x|^2)}{\rho(x)} &= \frac{\exp(-a|x|^2)}{\rho(x)} \frac{(G(1, \cdot) * \rho(\cdot))(x)}{(G(1, \cdot) * \rho(\cdot))(x)} \\ &\leq \left(\sup_{x \in \mathbb{R}^d, t \in [0,1]} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)} \right) \frac{\exp(-a|x|^2)}{(G(1, \cdot) * \rho(\cdot))(x)} \\ &\leq C_\rho(1) \frac{\exp(-a|x|^2)}{(G(1, \cdot) * \rho(\cdot))(x)}. \end{aligned}$$

Noticing that for some constant $\Theta > 0$, $\rho(x) \geq \Theta \prod_{\ell=1}^d \mathbb{I}_{[-1,1]}(x_\ell)$, a simple exercise shows that

$$C_{\rho,a} := \sup_{x \in \mathbb{R}^d} \frac{\exp(-a|x|^2)}{(G(1, \cdot) * \rho(\cdot))(x)} \leq \frac{1}{\Theta} \left(\sup_{x \in \mathbb{R}} \frac{\exp(-ax^2)}{(G(1, \cdot) * \mathbb{I}_{[-1,1]}(\cdot))(x)} \right)^d < \infty.$$

This proves the lemma with $K_{\rho,a} = C_\rho(1)C_{\rho,a}$. □

T: Solve

Theorem 2.5 (Theorem 1.2 [chen.huang:19:comparison]). *Suppose that*

(i) *The initial measure μ must satisfy the following integrability condition:*

$$\int_{\mathbb{R}^d} \exp(-a|x|^2) \|\mu(dx)\|_2 < \infty \quad \text{for all } a > 0. \quad (2.5) \quad \text{A:icon}$$

(ii) *The spectral measure $\widehat{f}$ satisfies Dalang's condition, namely,*

$$\Upsilon(\beta) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{\beta + |\xi|^2} < \infty \quad \text{for some and therefore any } \beta > 0. \quad (2.6) \quad \text{A:dalcon}$$

then (1.1) has a unique random field solution starting from μ . Moreover, the solution is $L^2(\Omega)$ continuous and is adapted to the filtration induced by W .

T: ExUn

Theorem 2.6 (Theorem 1.7 [chen.huang:19:comparison]). *Under the assumptions of Theorem 2.5, for any $t > 0$ and $x \in \mathbb{R}^d$, the solution to (1.1), $u(t, x)$, given by (1.3) is in $L^p(\Omega)$ and for $p \geq 2$*

$$\|u(t, x)\|_p \leq [\bar{\varsigma} + \sqrt{2}(G(t, \cdot) * \|\mu\|_p)(x)] H(t; \gamma_p)^{1/2}, \quad (2.7) \quad \text{E:mb}$$

where $\bar{\varsigma} = L_0/L_b$, $\gamma_p = 32pL_b^2$ and the function $H(t; \gamma_p)$ is nondecreasing in t (see [chen.huang:19:comparison] for the concrete definition of H). Moreover, if we assume that

$$\Upsilon(0) < \infty \quad \text{and} \quad \gamma_p < \frac{1}{4\Upsilon(0)}, \quad (2.8) \quad \text{A:dalcon0}$$

then this function $H(t; \gamma_p)$ is bounded in t . Hence, by choosing $C = (\sqrt{2} \vee \bar{\varsigma}) \sup_{t \geq 0} H(t; \gamma_p)^{1/2} < \infty$, we have that

$$\|u(t, x)\|_p \leq C(L_0 + J_p(t, x)), \quad (2.9) \quad \text{E:mb0}$$

where L_0 is from (1.4).

The following factorization property of the heat kernel will be used in this paper:

$$G(s - r, y - z)G(r, z - \sigma) = G(s, y - \sigma)G\left(\frac{r(s - r)}{s}, z - \sigma - \frac{r}{s}(y - \sigma)\right). \quad (2.10) \quad \text{E:gsim}$$

3 Proof of part (a) of Theorem 1.4

S:mainproof

For $\alpha \in (0, 1)$, $t > 0$ and $x \in \mathbb{R}^d$, define formally

$$\begin{aligned} (F_\alpha f)(t, x) &:= \int_0^t \int_{\mathbb{R}^d} (t - s)^{\alpha-1} G(t - s, x - y) f(s, y) \, ds dy \quad \text{and} \\ (Y_\alpha f)(t, x) &:= \int_0^t \int_{\mathbb{R}^d} (t - s)^{-\alpha} G(t - s, x - y) f(s, y) W(ds, dy). \end{aligned} \quad (3.1) \quad \text{E:Ops}$$

For F_α , the Step 1 of the proof of [tessitore.zabczyk:98:invariant] showed the following proposition:

P:F-Comp

Proposition 3.1. *Let ρ and $\hat{\rho}$ be given in condition (i) of Theorem 1.4 (see (1.18)). For any $q > 2$, $t_0 > 0$ and $\alpha \in (q^{-1}, 2^{-1})$, the operator F_α as an operator from $L^q((0, t_0); L_{\hat{\rho}}^2)$ to $L_{\rho}^2(\mathbb{R}^d)$ is compact.*

As for Y_α , we have the following result:

L:yineq

Lemma 3.2. *For any $\alpha \in (0, 1/2)$, denote (with slightly abuse of notation)*

$$Y(s, y) := [Y_\alpha b(\circ, u(\cdot, \circ))](s, y) = \int_0^s \int_{\mathbb{R}^d} (s - r)^{-\alpha} G(s - r, y - z) b(z, u(r, z)) W(dr, dz). \quad (3.2) \quad \text{E:Y}$$

Under the assumptions of Theorem 1.3 and (iii) of Theorem 1.4, then for all $q \geq 2$, $s > 0$ and $y \in \mathbb{R}^d$,

$$\|Y(s, y)\|_q^2 \leq C_q [J_q^*(s, y)]^2 \mathcal{H}_\alpha(s). \quad (3.3) \quad \text{E:NormY}$$

Moreover, if

(iv') for some $q \geq 2$ and $\alpha \in (0, 1/2)$, the weight function ρ (not necessarily admissible)

$$\int_0^t [\mathcal{G}_{\rho, q}(s) \mathcal{H}_\alpha(s)]^{q/2} ds < \infty, \quad \text{for all } t > 0, \quad (3.4) \quad \text{E:GH-int'}$$

then there exists a constant $\Theta = \Theta(q, L_b, L_0, \alpha)$ such that

$$\mathbb{E} \left(\int_0^t \|Y(s, \cdot)\|_\rho^q ds \right) \leq \Theta \int_0^t [\mathcal{G}_{\rho, q}(s) \mathcal{H}_\alpha(s)]^{q/2} ds < \infty. \quad (3.5) \quad \text{E:yineq}$$

Proof. In the proof, we use C to denote a generic constant that may change its value at each appearance. First, an application of Minkowski's inequality shows that

$$\mathbb{E} \left(\|Y(s, \cdot)\|_\rho^q \right) = \left\| \int_{\mathbb{R}^d} Y(s, y)^2 \rho(y) dy \right\|_{q/2}^{q/2} \leq \left(\int_{\mathbb{R}^d} \|Y(s, y)\|_q^2 \rho(y) dy \right)^{q/2}. \quad (3.6) \quad \boxed{\text{E:EYqr}}$$

Now we look to find an appropriate upper bound for $\|Y(s, y)\|_q^2$. By the Burkholder-Davis-Gundy inequality, as well as Minkowski's integral inequality to see that

$$\begin{aligned} \|Y(s, y)\|_q^2 &\leq C_q \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) \|b(z_1, u(r, z_1))\|_q f(z_1-z_2) \\ &\quad \times G(s-r, y-z_2) \|b(z_2, u(r, z_2))\|_q. \end{aligned}$$

Note that for the Lipschitz condition in (1.4), we have that

$$|b(x, u)| \leq |b(x, u) - b(x, 0)| + |b(x, 0)| \leq L_b |u| + L_0 \leq C(1 + |u|), \quad C := L_b \vee L_0.$$

We apply this and the moment bound (2.9) to $\|b(z_i, u(r, z_i))\|_q$ above to see that

$$\|b(z_i, u(r, z_i))\|_q \leq C \left(1 + \|u(r, z_i)\|_q \right) \leq C \left(1 + (G(r, \cdot) * \|\mu\|_q)(z_i) \right) = C J_q^*(r, z_i), \quad i = 1, 2, \quad (3.7) \quad \boxed{\text{E:NormB}}$$

where J_q^* is defined in (1.13). Therefore,

$$\begin{aligned} \|Y(s, y)\|_q^2 &\leq C \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1-z_2) \prod_{i=1}^2 \left(G(s-r, y-z_i) J_q^*(r, z_i) \right) \\ &= C \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu_q^*(d\sigma_1) \mu_q^*(d\sigma_2) \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\ &\quad \times f(z_1-z_2) \prod_{i=1}^2 \left(G(s-r, y-z_i) G(r, z_i-\sigma_i) \right) \\ &= C \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu_q^*(d\sigma_1) \mu_q^*(d\sigma_2) G(s, y-\sigma_1) G(s, y-\sigma_2) \\ &\quad \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1-z_2) \prod_{i=1}^2 G\left(\frac{r(s-r)}{s}, z_i-\sigma_i - \frac{r}{s}(y-\sigma_i)\right) \\ &\leq C(2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu_q^*(d\sigma_1) \mu_q^*(d\sigma_2) G(s, y-\sigma_1) G(s, y-\sigma_2) \\ &\quad \times \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right), \end{aligned}$$

where we have applied (2.10) and Plancherel's theorem. Therefore,

$$\|Y(s, y)\|_q^2 \leq C(2\pi)^{-2d} [J_q^*(s, y)]^2 \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right). \quad (3.8) \quad \boxed{\text{E_:NormY}}$$

By symmetry of $r(s-r)/s$ term and the fact that $r(s-r)/s \geq r/2$ for all $r \in [0, s/2]$, we see that the above double integral is bounded by

$$\leq 2 \int_0^{s/2} dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r}{2} |\xi|^2\right)$$

$$\begin{aligned}
&= 2^{2(1-\alpha)} \int_0^{s/4} dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r|\xi|^2) \\
&\leq 2^{2(1-\alpha)} \int_0^s dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r|\xi|^2) \\
&= 2^{2(1-\alpha)} \mathcal{H}_\alpha(s).
\end{aligned}$$

Plugging the above bound back to (3.8) proves (3.3).

It remains to prove (3.5). By the definition of $\mathcal{G}_\rho(t)$ in (1.15) and by (3.3), we see that

$$\int_{\mathbb{R}^d} \|Y(s, y)\|_q^2 \rho(y) dy \leq C \mathcal{G}_{\rho, q}(s) \mathcal{H}_\alpha(s).$$

Plugging the above expression to the far right-hand of (3.6) proves (3.5). Finally, the finiteness of the upper bound in (3.5) is guaranteed by condition (3.4). This completes the proof of Lemma 3.2. $\square$

L:Factorize

Lemma 3.3 (Factorization lemma). *Let $u(t, x)$ be the solution to (1.1) and Y be given in (3.2). Under the conditions of Theorem 1.3 and (iii) of Theorem 1.4, for all $\alpha \in (0, 1/2)$, the following factorization holds*

$$\frac{\sin(\alpha\pi)}{\pi} \int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y(s, \cdot)](x) ds = \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(z, u(r, z)) W(dr, dz).$$

As a consequence,

$$u(t, x) = J_0(t, x) + \frac{\sin(\alpha\pi)}{\pi} [F_\alpha Y](t, x), \quad \text{for all } t > 0 \text{ and } x \in \mathbb{R}^d. \quad (3.9)$$

E:1fac

Proof. The lemma is straightforward provided that one can switch the orders of stochastic and ordinary integrals:

$$\begin{aligned}
&\int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y(s, \cdot)](x) ds \\
&= \int_0^t ds (t-s)^{\alpha-1} \int_{\mathbb{R}^d} dy G(t-s, x-y) \\
&\quad \times \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) b(z, u(r, z)) W(dr, dz) \\
&= \int_0^t ds (t-s)^{\alpha-1} \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(t-r, x-z) b(z, u(r, z)) W(dr, dz) \quad (3.10)
\end{aligned}$$

E:Fubini_1

$$\begin{aligned}
&= \int_0^t \int_{\mathbb{R}^d} W(dr, dz) G(t-r, x-z) b(u(r, z)) \int_r^t ds (s-r)^{-\alpha} (t-s)^{\alpha-1} \quad (3.11) \\
&= \frac{\pi}{\sin(\alpha\pi)} \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(u(r, z)) W(dr, dz),
\end{aligned}$$

E:Fubini_2

where the last step is the Beta integral which requires that $\alpha \in (0, 1)$. It remains to justify the two applications of the stochastic Fubini's theorem (see Theorem 5.30 of Chapter one in [dalang.khoshnevisan.ea:09:minicourse], or also [walsh:86:introduction] or Theorem 4.18 of [da-prato.zabczyk:92:stochastic]) in (3.10) and (3.11) in the following two steps.

Step 1. In this step, we justify the change of orders in (3.10). Note that t, x and s are fixed. It suffices to prove the following condition:

$$I_1 := \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2$$

$$\begin{aligned}
& \times f(z_1 - z_2) \left(\prod_{i=1}^2 G(s - r, y - z_i) \right) \mathbb{E} \left(\prod_{i=1}^2 b(z_i, u(r, z_i)) \right) \\
& = \int_{\mathbb{R}^d} dy G(t - s, x - y) \|Y(s, y)\|_2^2 \\
& < +\infty.
\end{aligned}$$

But this follows immediately from (3.3). Indeed

$$\begin{aligned}
\int_{\mathbb{R}^d} dy G(t - s, x - y) \|Y(s, y)\|_2^2 & \leq \int_{\mathbb{R}^d} dy G(t - s, x - y) (J_2^*(s, y))^2 \mathcal{H}_\alpha(s) \\
& = C \mathcal{H}_\alpha(s) \int_{\mathbb{R}^d} dy G(t - s, x - y) (J_2^*(s, y))^2.
\end{aligned}$$

Then Condition (1.15) implies that $I_1 < \infty$ as $\rho(y)$ cannot decay as fast as $y \mapsto G(t - s, x - y)$ (See Remark 2.2).

Step 2. Similarly, as for (3.11), we need to show that

$$\begin{aligned}
I_2 & := \int_0^t ds (t - s)^{\alpha-1} \int_0^s dr (s - r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\
& \quad \times f(z_1 - z_2) \left(\prod_{i=1}^2 G(t - r, x - z_i) \right) \mathbb{E} \left(\prod_{i=1}^2 b(z_i, u(r, z_i)) \right) < \infty.
\end{aligned}$$

By the Cauchy Schwartz inequality, (3.7) and because $\alpha \in (0, 1/2)$,

$$\begin{aligned}
I_2 & \leq C \int_0^t ds (t - s)^{\alpha-1} \int_0^s dr (s - r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left(G(t - r, x - z_i) J_2^*(r, z_i) \right) \\
& = C' \int_0^t dr (t - r)^{-\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left(G(t - r, x - z_i) J_2^*(r, z_i) \right).
\end{aligned}$$

Now by the same arguments as those leading to (3.3) (with 2α there replaced by α), we see that

$$I_2 \leq C J_*^2(t, x) \int_0^t dr r^{-\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp \left(-\frac{r(t-r)}{r} |\xi|^2 \right),$$

which is finite by (1.20) where we replace α with $\alpha/2$. Indeed, since $r(t-r) \geq tr/2$ for $r \in [0, t]$, then

$$\int_0^t dr r^{-\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp \left(-\frac{r(t-r)}{r} |\xi|^2 \right) \leq \int_0^t dr r^{-\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp \left(-\frac{r}{2} |\xi|^2 \right) < \infty.$$

This completes the proof of Lemma 3.3. $\square$

We are now ready to prove part (b) of Theorem 1.4.

Proof of Theorem 1.4 (b). Recall that from Lemma 3.3 implies that for any $t_0 > 0$,

$$u(t_0, x) = \left(G(t_0, \cdot) * u(0, \cdot) \right)(x) + \frac{\sin(\alpha\pi)}{\pi} [F_\alpha Y](t_0, x),$$

and again recall that we assume $u(0, \cdot)$ satisfies (1.10). Now define the set $\mathcal{K}(\Lambda)$ as

$$\mathcal{K}(\Lambda) = \{ (G(t_0, \cdot) * y(\cdot))(x) + (F_\alpha h)(x), \quad \text{where } \|y\|_{\dot{\rho}} \leq \Lambda \text{ and } \|h\|_{L^q(0,1,L^2_{\dot{\rho}})} \leq \Lambda \}.$$

Then by Proposition 3.1, $\mathcal{K}(\Lambda)$ is relatively compact in L^2_ρ for $\Lambda > 0$. Moreover, by (3.5) and Chebyshev's inequality we have

$$\begin{aligned} \mathbb{P}\{u(t_0, \cdot) \notin \mathcal{K}(\Lambda)\} &\leq \mathbb{P}\left[\left(\int_0^{t_0} \|Y(s, \cdot)\|_{\hat{\rho}}^q ds\right)^{1/q} > \frac{\pi\Lambda}{\sin(\alpha\pi)}\right] + \mathbb{P}\left[\|u(0, \cdot)\|_{\hat{\rho}} > \Lambda\right] \\ &\leq \frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \mathbb{E} \int_0^{t_0} \|Y(s, \cdot)\|_{\hat{\rho}}^q ds + \frac{1}{\Lambda^2} \mathbb{E} \left(\|u(0, \cdot)\|_{\hat{\rho}}^2\right) \\ &\leq \frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \Theta \int_0^{t_0} (\mathcal{G}_{\hat{\rho}, q}(s) \mathcal{H}_\alpha(s))^{q/2} ds + \frac{1}{\Lambda^2} \mathbb{E} \left(\|u(0, \cdot)\|_{\hat{\rho}}^2\right) < \infty \end{aligned} \quad (3.12) \quad \boxed{\text{E:Restartbd}}$$

With all this said, (3.12) implies that

$$\mathbb{P}\{u(t_0, \cdot) \in \mathcal{K}(\Lambda)\} \geq 1 - \frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \left(\Theta \int_0^{t_0} (\mathcal{G}_{\hat{\rho}, q}(s) \mathcal{H}_\alpha(s))^{q/2} ds \right) - \frac{1}{\Lambda^2} \mathbb{E} \left(\|u(0, \cdot)\|_{\hat{\rho}}^2\right)$$

Therefore, we can choose $\Lambda > 0$ big enough such that

$$\frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \left(\Theta \int_0^{t_0} (\mathcal{G}_{\hat{\rho}, q}(s) \mathcal{H}_\alpha(s))^{q/2} ds \right) + \frac{1}{\Lambda^2} \mathbb{E} \left(\|u(0, \cdot)\|_{\hat{\rho}}^2\right) < \epsilon,$$

thus we get that

$$\mathbb{P}\{u(t_0, \cdot) \in \mathcal{K}(\Lambda)\} \geq 1 - \epsilon.$$

We can now deduce from this and the semigroup property of the solution to the heat equation that for any $t > t_0$,

$$\mathbb{P}\{u(t, \cdot) \in \mathcal{K}(\Lambda)\} \geq (1 - \epsilon) \mathbb{P}\{\|u(t - t_0, \cdot)\|_{\hat{\rho}} < \Lambda\}.$$

We now apply Chebyshev's inequality and (1.19) to see that for all $t > t_0$ that there exists a $C > 0$ such that

$$\mathbb{P}\{\|u(t - t_0, \cdot)\|_{\hat{\rho}} \geq \Lambda\} \leq \frac{\mathbb{E} \left(\|u(t - t_0, \cdot)\|_{\hat{\rho}}^2\right)}{\Lambda^2} \leq C \frac{\mathcal{G}_{\hat{\rho}, q}(t - t_0)}{\Lambda^2} \leq \frac{C}{\Lambda^2}.$$

Thus, by enlarging Λ if necessary,

$$\mathbb{P}\{\|u(t - t_0, \cdot)\|_{\hat{\rho}} \geq \Lambda\} \leq \epsilon, \quad \forall t \geq t_0,$$

thus completing the proof of (1.22). □